

I. INTRODUCTION (1)² Risk management principles are effectively utilized in many areas of business and government including finance, insurance, occupational safety, public health, pharmacovigilance, and by agencies regulating these industries. In the pharmaceutical sector, the principles and framework of ICH Q9, coupled with the official ICH training material that supports this guideline, are instrumental in enhancing the application of effective quality risk management by industry and regulators. The importance of quality systems has been recognized in the pharmaceutical industry and it is evident that quality risk management is a valuable component of an effective quality system. It is commonly understood that risk is defined as the combination of the probability of occurrence of harm and the severity of that harm. However, achieving a shared understanding of the application of risk management among diverse stakeholders is difficult because each stakeholder might perceive different potential harms, place a different probability on each harm occurring and attribute different severities to each harm. In addition, subjectivity can directly impact the effectiveness of risk management activities and the decisions made. Therefore, it is important that subjectivity is managed and minimized. In relation to pharmaceuticals, although there are a variety of stakeholders, including patients and medical practitioners as well as government and industry, the protection of the patient is of prime importance when managing the risk to product quality and availability, when availability risks arise from quality/manufacturing issues. The manufacturing and use of a drug (medicinal) product, including its components, necessarily entail some degree of risk. The risk to its quality is just one component of the overall risk. It is important to understand that product quality is assured based on appropriate risk- based decision 1This guidance was developed within the Expert Working Group (Quality) of the International Council for Harmonisation of Technical Requirements for Registration of Pharmaceuticals for Human Use (ICH) and has been subject to consultation by the regulatory parties, in accordance with the ICH process. This document has been endorsed by the ICH Assembly at Step 4 of the ICH process, January 2023. At Step 4 of the process, the final draft is recommended for adoption to the regulatory bodies of the ICH regions. 2The numbers in parentheses reflect the organizational breakdown of the document endorsed by the ICH Assembly at Step 4 of the ICH process, January 2023. making throughout the product lifecycle, such that the attributes that are important to the quality of the drug (medicinal) product are maintained and the product remains safe and effective. An effective quality risk management approach can further ensure the quality of the drug (medicinal) product to the patient by providing a proactive means to identify and control potential quality issues during development, manufacturing, and distribution. This includes an appropriate application of root cause analysis that can identify and address the root cause(s) and other causal factors (e.g., human-related) of such issues. A proactive approach to quality risk management is beneficial, as it facilitates robust product design and continual

improvement, and it is of strategic importance in achieving an effective pharmaceutical quality system (see the International Council for Harmonisation (ICH) guidance for industry Q10 Pharmaceutical Quality System (ICH Q10) (April 2009)³ for guidance in relation to an effective pharmaceutical quality system). Additionally, use of quality risk management can improve the decision-making if a quality problem arises. In the development phase and as part of validation, quality risk management is part of building knowledge and understanding risk scenarios, so that appropriate risk control can be decided upon for use during the commercial manufacturing phase. In this context, knowledge is used to make informed risk-based decisions, trigger re-evaluations and stimulate continual improvements. Effective and proactive quality risk management can enable better, more informed, and timely decisions throughout the lifecycle. This can provide regulators with greater assurance of a company's ability to deal with potential risks and avert problems and can beneficially affect the extent and level of direct regulatory oversight. The application of digitalization and emerging technologies in the manufacture and control of drug (medicinal) products can lead to risk reduction when such technologies are fit for their intended use. However, they can also introduce other risks that may need to be controlled. The application of quality risk management to the design, validation and technology transfer of advanced production processes and analytical methods, advanced data analysis methods and computerized systems is important. The purpose of this document is to offer a systematic approach to quality risk management that leads to better, more informed, and timely decisions. It serves as a foundation or resource document that is independent of, yet supports, other ICH Quality documents and complements existing quality practices, requirements, standards, and guidelines within the pharmaceutical industry and regulatory environment. It specifically provides guidance on the principles and some of the tools of quality risk management that can enable more effective and consistent risk based decisions, both by regulators and industry, regarding the quality of drug substances and drug (medicinal) products across the product lifecycle. It is not intended to create any new expectations beyond the current regulatory requirements. An understanding of formality in quality risk management may lead to resources being used more efficiently, where lower risk issues are dealt with via less formal means, freeing up resources for managing higher risk issues and more complex problems that may require increased levels of rigor and effort. An understanding of formality can also support risk-based decision-making, where the level of formality that is applied may reflect the degree of importance of the decision, as well as the level of uncertainty and complexity which may be present. Appropriate use of quality risk management can facilitate but does not obviate industry's obligation to comply with regulatory requirements and does not replace appropriate communications between industry and regulators. Quality risk management should not be used in a manner where

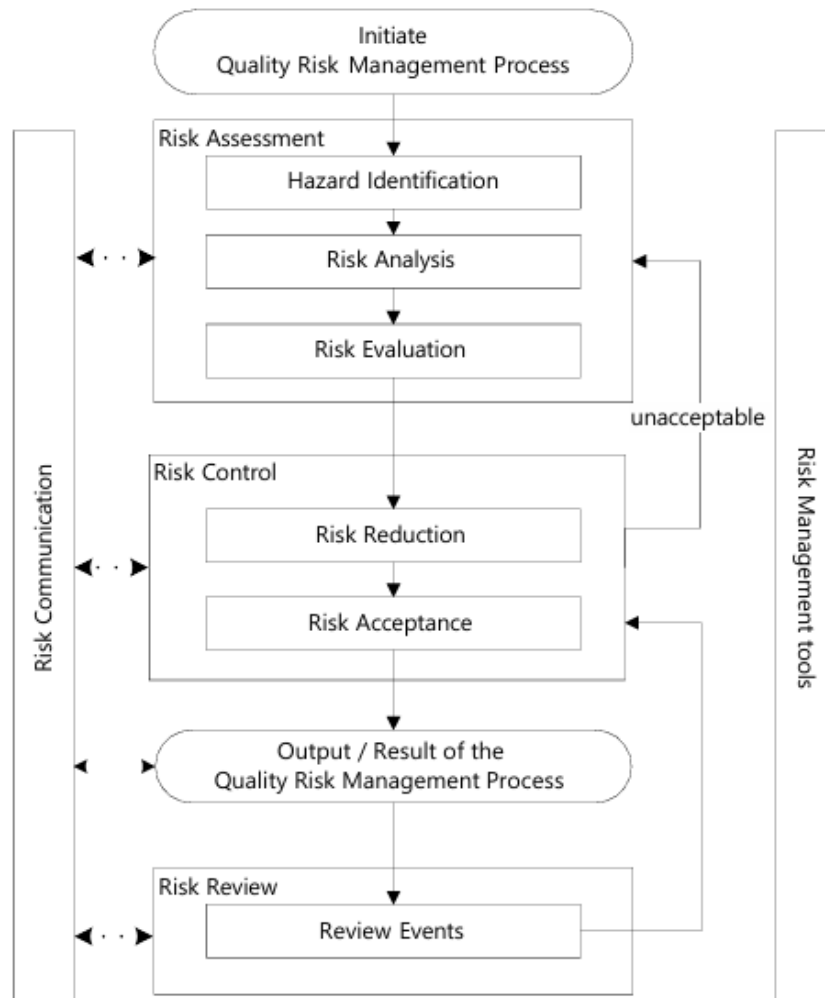
decisions are made that justify a practice that would otherwise, in accordance with regulations and/or guidance, be deemed unacceptable. In general, FDA's guidance documents do not establish legally enforceable responsibilities. Instead, guidances describe the Agency's current thinking on a topic and should be viewed only as recommendations, unless specific regulatory or statutory requirements are cited. The use of the word should in Agency guidances means that something is suggested or recommended, but not required.

II. SCOPE (2) This guideline provides principles and examples of tools for quality risk management that can be applied to different aspects of pharmaceutical quality. These aspects include development, manufacturing, distribution, and the inspection and submission/review processes throughout the lifecycle of drug substances, drug (medicinal) products, biological and biotechnological products (including the use of raw materials, solvents, excipients, packaging and labeling materials in drug (medicinal) products, biological and biotechnological products).

III. PRINCIPLES OF QUALITY RISK MANAGEMENT (3) Two primary principles of quality risk management are: The evaluation of the risk to quality should be based on scientific knowledge and ultimately link to the protection of the patient. (Note: Risk to quality includes situations where product availability may be impacted, leading to potential patient harm.) The level of effort, formality and documentation of the quality risk management process should be commensurate with the level of risk.

IV. GENERAL QUALITY RISK MANAGEMENT PROCESS (4) Quality risk management is a systematic process for the assessment, control, communication, and review of risks to the quality of the drug (medicinal) product across the product lifecycle. A model for quality risk management is outlined in the diagram (Figure 1). Other models could be used. The emphasis on each component of the framework might differ from case to case but a robust process will incorporate consideration of all the elements at a level of detail that is commensurate with the specific risk.

Figure 1: Overview of a typical quality risk management process.



A. Responsibilities (4.1) Quality risk management activities are usually, but not always, undertaken by interdisciplinary teams. When teams are formed, they should include experts from the appropriate areas (e.g., quality unit, product development, business development, engineering, regulatory affairs, production operations, sales and marketing, supply chain, legal, statistics and clinical) in addition to individuals who are knowledgeable about the quality risk management process.

Decision makers should:

- Take responsibility for coordinating quality risk management across various functions and departments of their organization;
- Assure that a quality risk management process is defined, deployed, and reviewed and that adequate resources and knowledge are available; and
- Assure that subjectivity in quality risk management

activities is managed and minimized, to facilitate scientifically robust risk-based decision-making

B. Initiating a Quality Risk Management Process (4.2) Quality risk management should include systematic processes designed to coordinate, facilitate and improve science-based decision-making with respect to risk. Possible steps used to initiate and plan a quality risk management process might include the following: • Define the problem and/or risk question, including pertinent assumptions identifying the potential for risk • Assemble background information and/ or data on the potential hazard, harm, or human health impact relevant to the risk assessment • Identify a leader and necessary resources • Specify a timeline, deliverables, and appropriate level of decision-making for the risk management process

C. Risk Assessment (4.3) Risk assessment consists of the identification of hazards and the analysis and evaluation of risks associated with exposure to those hazards (as defined below). Quality risk assessments begin with a well-defined problem description or risk question. When the risk in question is well defined, an appropriate risk management tool (see examples in section V (5)) and the types of information needed to address the risk question will be more readily identifiable. As an aid to clearly defining the risk(s) for risk assessment purposes, three fundamental questions are often helpful: (1) What might go wrong? (2) What is the likelihood (probability) it will go wrong? (3) What are the consequences (severity)? Hazard identification is a systematic use of information to identify hazards referring to the risk question or problem description. Information can include historical data, theoretical analysis, informed opinions, and the concerns of stakeholders. Hazard identification addresses the “What might go wrong?” question, including identifying the possible consequences. This provides the basis for further steps in the quality risk management process. Risk analysis is the estimation of the risk associated with the identified hazards. It is the qualitative or quantitative process of linking the likelihood of occurrence and severity of harms. In some risk management tools, the ability to detect the harm (detectability) also factors in the estimation of risk. Risk evaluation compares the identified and analyzed risk against given risk criteria. Risk evaluations consider the strength of evidence for all three of the fundamental questions. In doing an effective risk assessment, the robustness of the data set is important because it determines the quality of the output. Revealing assumptions and reasonable sources of uncertainty will enhance confidence in this output and/or help identify its limitations. Uncertainty is due to a combination of incomplete knowledge about a process and its expected or unexpected variability. Typical sources of uncertainty include gaps in knowledge, gaps in pharmaceutical science and process understanding, sources of harm (e.g., failure modes of a process, sources of variability), and probability of detection of

problems. The output of a risk assessment is either a quantitative estimate of risk or a qualitative description of a range of risk. When risk is expressed quantitatively, a numerical probability is used. Alternatively, risk can be expressed using qualitative descriptors, such as “high,” “medium,” or “low,” which should be defined in as much detail as possible. Sometimes a “risk score” is used to further define descriptors in risk ranking. In quantitative risk assessments, a risk estimate provides the likelihood of a specific consequence, given a set of risk-generating circumstances. Thus, quantitative risk estimation is useful for one particular consequence at a time. Alternatively, some risk management tools use a relative risk measure to combine multiple levels of severity and probability into an overall estimate of relative risk. The intermediate steps within a scoring process can sometimes employ quantitative risk estimation.

D. Risk Control (4.4) Risk control includes decision-making to reduce and/or accept risks. The purpose of risk control is to reduce the risk to an acceptable level. The amount of effort used for risk control should be proportional to the significance of the risk. Decision makers might use different processes, including benefit-cost analysis, for understanding the optimal level of risk control. Risk control might focus on the following questions: (1) Is the risk above an acceptable level? (2) What can be done to reduce or eliminate risks? (3) What is the appropriate balance among benefits, risks, and resources? (4) Are new risks introduced as a result of the identified risks being controlled?

Risk reduction focuses on processes for mitigation or avoidance of quality risk when it exceeds a specified (acceptable) level (see Fig. 1). Risk reduction might include actions taken to mitigate the severity and probability of harm. Processes that improve the detectability of hazards and quality risks might also be used as part of a risk control strategy. The implementation of risk reduction measures can introduce new risks into the system or increase the significance of other existing risks. Hence, it might be appropriate to revisit the risk assessment to identify and evaluate any possible change in risk after implementing a risk reduction process.

Risk acceptance is a decision to accept risk. For some types of harms, even the best quality risk management practices might not entirely eliminate risk. In these circumstances, it might be agreed that an appropriate quality risk management strategy has been applied and that quality risk is reduced to a specified (acceptable) level. This (specified) acceptable level will depend on many parameters and should be decided on a case-by-case basis.

ANNEX I: QUALITY RISK MANAGEMENT METHODS AND TOOLS

The purpose of this annex is to provide a general overview of and references for some of the primary tools that might be used in quality risk management by industry and regulators. The references are included as an aid to gain more knowledge and detail about the particular tool. This is not an exhaustive list. It is important to note that no one tool or set of tools is applicable to every situation in which a quality risk management procedure is used. It is neither always appropriate nor always necessary to use highly formal quality risk management methods and tools. The use of less formal quality risk management methods and tools can also be considered acceptable. See section V.A (5.1) in the main body of this guidance for guidance on what constitutes formality in quality risk management.

A. Basic Risk Management Facilitation Methods (I.1) Some of the simple techniques that are commonly used to structure risk management by organizing data and facilitating decision-making are: • Flowcharts • Check Sheets • Process Mapping • Cause and Effect Diagrams (also called an Ishikawa diagram or fish bone diagram)

B. Failure Mode Effects Analysis (FMEA) (I.2) FMEA (see IEC 60812) provides for an evaluation of potential failure modes for processes and their likely effect on outcomes and/or product performance. Once failure modes are established, risk reduction can be used to eliminate, contain, reduce, or control the potential failures. FMEA relies on product and process understanding. FMEA methodically breaks down the analysis of complex processes into manageable steps. It is a powerful tool for summarizing the important modes of failure, factors causing these failures and the likely effects of these failures. Potential Areas of Use(s) FMEA can be used to prioritize risks and monitor the effectiveness of risk control activities. FMEA can be applied to equipment and facilities and might be used to analyze a manufacturing operation and its effect on product or process. It identifies elements/operations within the system that render it vulnerable. The output/ results of FMEA can be used as a basis for design or further analysis or to guide resource deployment.

C. Failure Mode, Effects, and Criticality Analysis (FMECA) (I.3) FMEA might be extended to incorporate an investigation of the degree of severity of the consequences, their respective probabilities of occurrence, and their detectability, thereby becoming a Failure Mode Effect and Criticality Analysis (FMECA; see IEC 60812). For such an analysis to be performed, the product or process specifications should be established. FMECA can identify places where additional preventive actions might be appropriate to minimize risks. Potential Areas of Use(s) FMECA application in the pharmaceutical industry should mostly be utilized for failures and risks associated with manufacturing processes; however, it is not limited to this application. The output of an FMECA is a relative risk “score” for each failure mode, which is used to rank the modes on a relative risk basis.

D. Fault Tree Analysis (FTA) (I.4) The FTA tool (see IEC 61025) is an approach that assumes failure of the functionality of a product or process. This tool evaluates system (or subsystem) failures one at a time but can combine multiple causes of failure by identifying causal chains. The results are represented pictorially in the form of a tree of fault modes. At each level in the tree, combinations of fault modes are described with logical operators (AND, OR, etc.). FTA relies on the experts' process understanding to identify causal factors. Potential Areas of Use(s) FTA can be used to establish the pathway to the root cause of the failure. FTA can be used to investigate complaints or deviations in order to fully understand their root cause and to ensure that intended improvements will fully resolve the issue and not lead to other issues (i.e., solve one problem yet cause a different problem). Fault Tree Analysis is an effective tool for evaluating how multiple factors affect a given issue. The output of an FTA includes a visual representation of failure modes. It is useful both for risk assessment and in developing monitoring programs.

E. Hazard Analysis and Critical Control Points (HACCP) (I.5) HACCP is a systematic, proactive, and preventive tool for assuring product quality, reliability, and safety (see WHO Technical Report Series No 908, 2003 Annex 7). It is a structured approach that applies technical and scientific principles to analyze, evaluate, prevent, and control the risk or adverse consequence(s) of hazard(s) due to the design, development, production, and use of products. HACCP consists of the following seven steps: 1. Conduct a hazard analysis and identify preventive measures for each step of the process 2. Determine the critical control points 3. Establish critical limits 4. Establish a system to monitor the critical control points 5. Establish the corrective action to be taken when monitoring indicates that the critical control points are not in a state of control 6. Establish system to verify that the HACCP system is working effectively 7. Establish a record-keeping system Potential Areas of Use(s) HACCP might be used to identify and manage risks associated with physical, chemical, and biological hazards (including microbiological contamination). HACCP is most useful when product and process understanding is sufficiently comprehensive to support identification of critical control points. The output of a HACCP analysis is risk management information that facilitates monitoring of critical points not only in the manufacturing process but also in other life cycle phases.

F. Hazard Operability Analysis (HAZOP) (I.6) HAZOP (see IEC 61882) is based on a theory that assumes that risk events are caused by deviations from the design or operating intentions. It is a systematic brainstorming technique for identifying hazards using so-called "guide-words." "Guide-words" (e.g., No, More, Other Than, Part of, etc.) are applied to relevant parameters (e.g., contamination, temperature) to help identify potential deviations from normal use or design intentions. It often uses a team of people with expertise covering the design of the process or product and its application. Potential Areas

of Use(s) HAZOP can be applied to manufacturing processes, including outsourced production and formulation as well as the upstream suppliers, equipment and facilities for drug substances and drug (medicinal) products. It has also been used primarily in the pharmaceutical industry for evaluating process safety hazards. As is the case with HACCP, the output of a HAZOP analysis is a list of critical operations for risk management. This facilitates regular monitoring of critical points in the manufacturing process.

G. Preliminary Hazard Analysis (PHA) (I.7) PHA is a tool of analysis based on applying prior experience or knowledge of a hazard or failure to identify future hazards, hazardous situations and events that might cause harm, as well as to estimate their probability of occurrence for a given activity, facility, product, or system. The tool consists of: (1) the identification of the possibilities that the risk event happens; (2) the qualitative evaluation of the extent of possible injury or damage to health that could result; (3) a relative ranking of the hazard using a combination of severity and likelihood of occurrence; and (4) the identification of possible remedial measures. Potential Areas of Use(s) PHA might be useful when analyzing existing systems or prioritizing hazards where circumstances prevent a more extensive technique from being used. It can be used for product, process, and facility design as well as to evaluate the types of hazards for the general product type, then the product class, and finally the specific product. PHA is most commonly used early in the development of a project when there is little information on design details or operating procedures; thus, it will often be a precursor to further studies. Typically, hazards identified in the PHA are further assessed with other risk management tools such as those in this section.

H. Risk Ranking and Filtering (I.8) Risk ranking and filtering is a tool for comparing and ranking risks. Risk ranking of complex systems typically requires evaluation of multiple diverse quantitative and qualitative factors for each risk. The tool involves breaking down a basic risk question into as many components as needed to capture factors involved in the risk. These factors are combined into a single relative risk score that can then be used for ranking risks. “Filters,” in the form of weighting factors or cut-offs for risk scores, can be used to scale or fit the risk ranking to management or policy objectives. Potential Areas of Use(s) Risk ranking and filtering can be used to prioritize manufacturing sites for inspection/audit by regulators or industry. Risk ranking methods are particularly helpful in situations in which the portfolio of risks and the underlying consequences to be managed are diverse and difficult to compare using a single tool. Risk ranking is useful when management needs to evaluate both quantitatively assessed and qualitatively assessed risks within the same organizational framework.

I. Supporting Statistical Tools (I.9) Statistical tools can support and facilitate quality risk management. They can enable effective data assessment, aid in determining the significance of the data set(s) and facilitate more reliable decision-making. A listing of some of the principal statistical tools commonly used in the pharmaceutical industry is provided: Control Charts, for example: • Acceptance Control Charts (see ISO 7870-3:2020) • Cumulative Sum Charts (see ISO 7870-4:2021) • Shewhart Control Charts (see ISO 7870-2:2013) • Weighted Moving Average • Design of Experiments (DOE) • Histograms • Pareto Charts Process Capability Analysis